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1. Introduction

The Boeing 767-200 ER (Extended Range) is an American medium to long range wide-body commercial aircraft that has been in service since 1984 as a long-range variant of the baseline 767-200. The aircraft was developed to serve long-haul routes, making it popular choice among international carriers seeking an efficient alternative to four-engine wide-body aircraft.



Figure 1: Boeing 767-200 ER [2]

The B767-200ER is powered by two high-bypass ratio General Electric CF6-80C2 turbofan engines. Unlike afterburning turbojets found in military application, CF6-80C2 engines are optimised for fuel efficiency over long distances. This is a critical design decision given that the aircraft has a design range of approximately 12200km.

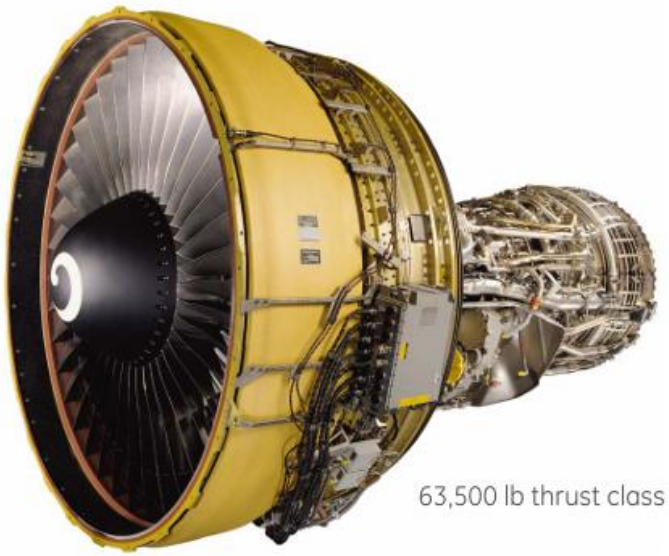


Figure 2: General Electric CF6-80C2 [3]

The wide-body fuselage of the aircraft necessitated large, high thrust engines capable of maintaining cruise altitude of up to 43000 ft, and the General Electric CF6-80C2 was able to meet that requirement.

2. Analysis

2.1. Assumptions

The following analysis was conducted under these assumptions:

- The working fluid is air and is assumed to an ideal gas.
- Constant air density at any given altitude.
- The velocity of air close to the inlet of the diffuser was approximated to be 0m/s.
- The diffuser and nozzle are isentropic.
- The velocity of air that runs through the engine is assumed to be close to zero.
- The exhaust pressure matches the atmospheric pressure.
- The compressor and turbine are not isentropic because of losses due to efficiencies.
- The combustor is isobaric.
- Changes in potential and kinetic energy are negligible inside the engine.
- Engine does not have preheaters, afterburners and intercoolers and thus, their effects were neglected.
- The turbine inlet temperature was assumed to be 1608 K.
- The 14-stage compressor does not have any reheaters or intercoolers between compressors, therefore, it was assumed to be a single compressor. The same modelling was done for the turbines.

2.2. Methodology

Engine parameters are provided in Table 4 located in the Appendix section of this report. of this report. An Iterative method was used to find optimum values of the compressor and turbine efficiencies. The efficiencies were estimated by guessing the initial values of efficiencies for static sea lev analysis and calculating the resultant thrust that would be compared to the published value. Upon determining suitable values, the efficiencies values are also used in the subsequent cruise condition analysis.

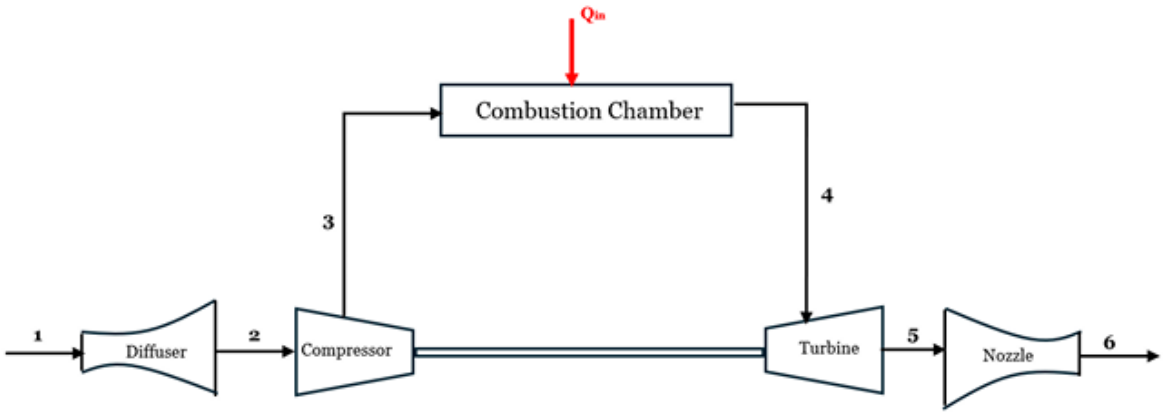


Figure 3: Brayton cycle for the CF6-80C2

3. Static Sea-Level analysis

To calibrate the unknown engine parameters (efficiencies of the compressor and turbine), a static sea-level analysis of the engine was performed with the inlet conditions set to standard sea-level temperature and pressure (288.15K and101.235 Kpa). The compressor and turbine efficiencies were initially assigned values of 0.8 and 0.9 respectively. These were deliberately chosen to reflect the least favourable operating conditions, which would correspond to a lowest possible exit velocity. This is because compressor efficiency and turbine efficiency have opposing effects. A high compressor efficiency would result in greater thrust and therefor a higher exit velocity, while a high turbine efficiency reduces the exit velocity.

Table 1: Static sea-level analysis (initial calculations)

No	Velocity	Pressure [kPa]	Pr	Temperature [K]	Enthalpy [KJ/Kg]
1	*0*	*101.325*	*1.204201*	*288.15*	*288.3026*
2	-	*101.325*	*1.204201*	*288.15*	*288.3026*
3s	-	*3080.28*	*31.7844894*	*718.5639597*	*762.9365718*
3	-	*3080.28*	*61.4407924*	*853.9919575*	*881.5950648*
4	-	*3080.28*	*808.36*	*1608*	*1767.342*
5	-	*536.5182124*	*174.146193*	*1111.184434*	*1174.049535*
5s	-	*536.5182124*	*140.798844*	*1054.148762*	*1108.12815*
6	*906.5862803*	*101.325*	*26.5907895*	*685.1443713*	*697.1788085*

The following set of equations were used to calculate the resultant thrust. The fan jet velocity was assumed to be 180m/s.

$$\dot{m}_{total} = \dot{m}_{core} + \dot{m}_{bypass}$$

$$BPR = \frac{\dot{m}_{bypass}}{\dot{m}_{core}}$$

$$\dot{m}_{core} = \frac{\dot{m}_{core}}{1 + BPR}$$

$$\dot{m}_{core} = \frac{BPR(\dot{m}_{total})}{1 + BPR}$$

$$F = \dot{m}_{core}(V_{exit} - V_0) + \dot{m}_{bypass}(V_{fan} - V_0)$$

Using this analysis the thrust was found to be 238466.33 N which found to be 2% higher than the published value (233531 N). The initially selected efficiencies were found to yield the best results.

4. Cruise Condition Analysis

The aircraft cruises at an altitude of 35000 ft and at speed of 252.07 m/s (Mach 0.85). The pressure and temperature were determined according to the U.S Standard Atmospheric and the pressure and temperature were found to be 23.84 Kpa and 218.8 K respectively. The efficiencies determined for static sea-level analysis were also used for cruise

Table 2: Cruise condition analysis

No	Velocity	Pressure [kPa]	Pr	Temperature [K]	Enthalpy [KJ/Kg]
1	*252.0711824*	*23.84*	*0.460564*	*218.8*	*218.77*
2	*-*	*38.20861822*	*0.73815076*	*250.4879885*	*250.5399405*
3s	*-*	*1161.541994*	*19.8534673*	*630.1683417*	*664.6928294*
3	*-*	*1161.541994*	*37.5159154*	*750.8641429*	*768.2310516*
4	*-*	*1161.541994*	*808.36*	*1608*	*1767.342*
5	*-*	*264.7026143*	*219.196726*	*1175.995635*	*1249.650889*
5s	*-*	*264.7026143*	*184.216331*	*1126.740794*	*1192.129654*
6	*1078.799075*	*23.84*	*16.5911370*	*603.0503632*	*610.2259317*

The calculated thrust (58492.0797 N) exceeds the published cruise thrust value (50398 N), which can be attributed to the influence of the incoming airflow on engine performance. As air enters the system, the velocity changes across the diffuser decreases, resulting in higher pressure and temperature (and enthalpy) at the engine inlet. As a consequence, this increases the mass flow rate, which is beneficial for the compressor as it gets more air to compress, thereby improving overall engine performance. Therefore, the compressor operates more efficiently under these conditions, and ideally this analysis would be repeated with revised efficiency values for the turbine and compressor to reflect this better. However, since the efficiencies were held constant throughout this analysis to allow for a direct comparison, the resulting exit velocity and thrust are higher than would otherwise be expected.

5. Summary of results

A summary of the results is given in Table 3 below.

Table 3: Summary of the result

	Thrust at Sea Level (N)	Thrust at Cruise (N)
Published	233531	50398
Hand Calculations	238466.33	58492.07984
Percentage error (%)	2.07	13.83

6. T-S diagrams

6.1. Static sea-level diagram

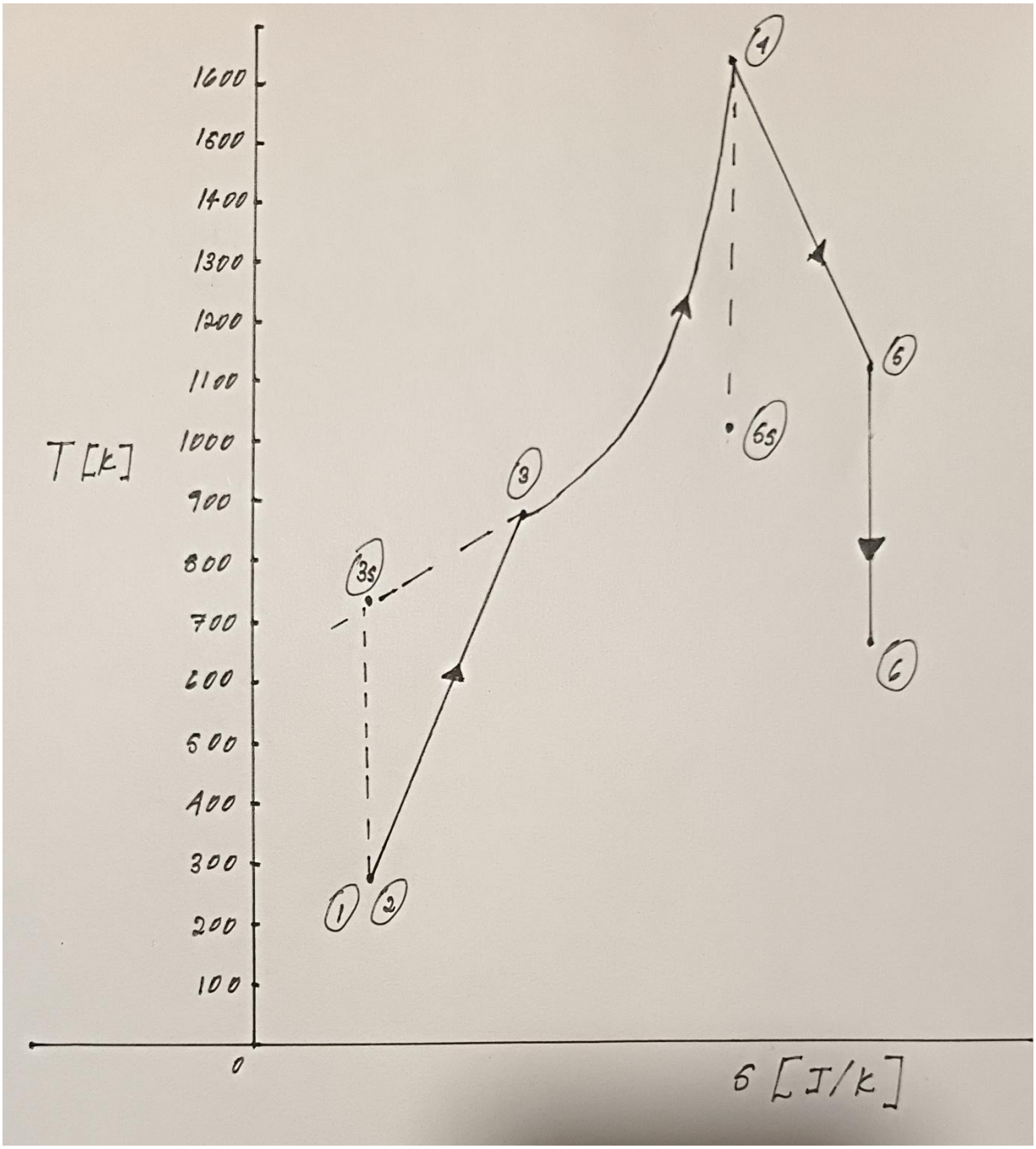


Figure 4: Static sea level T-S diagram

6.2. Cruise condition diagram

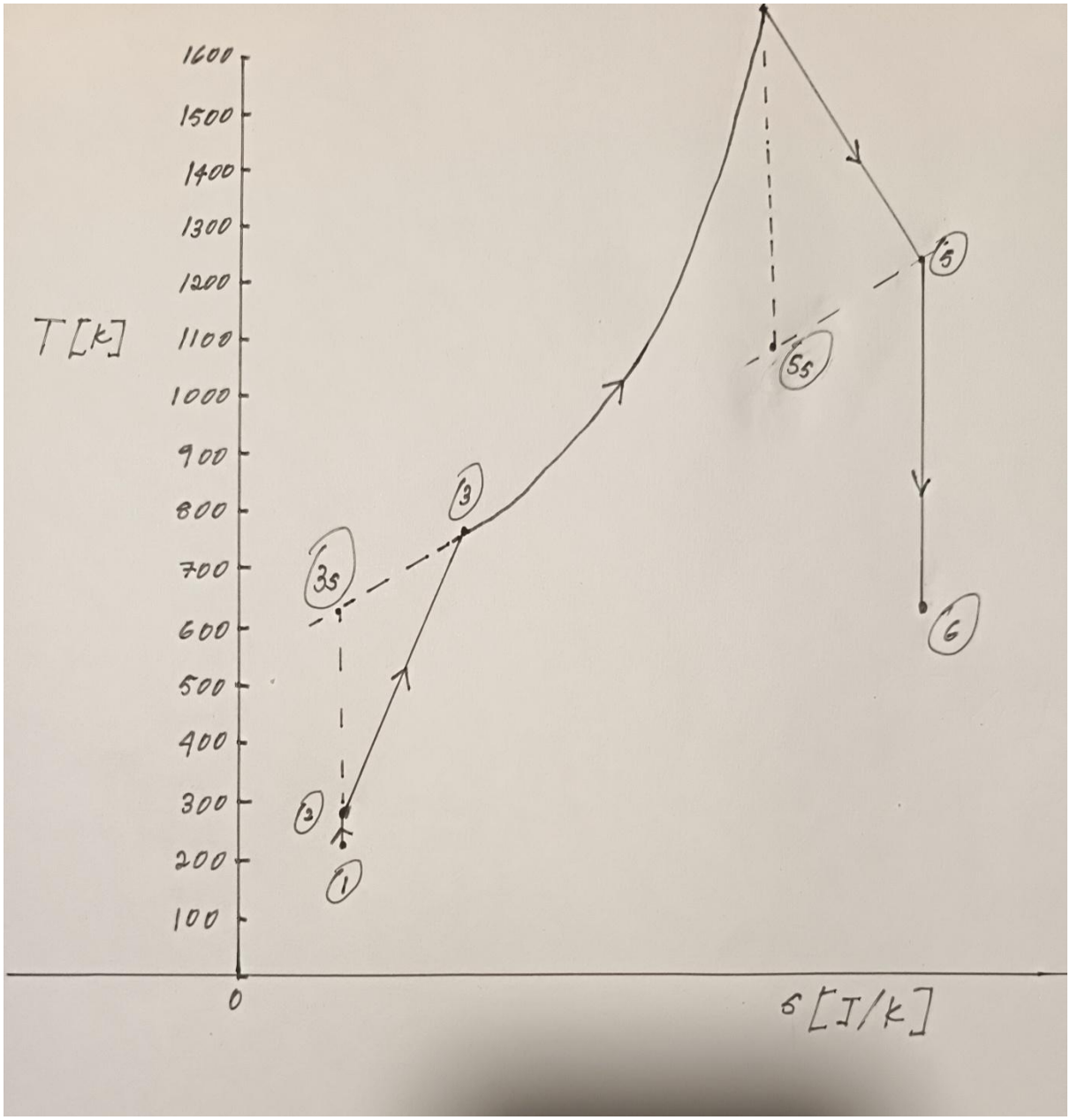


Figure 5: Cruise T-s diagram

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[1] Roux, E. (2007) *Turbofan and turbojet engines: Database handbook*. Blagnac: Élodie Roux.

[2] 767 *3D models download - free3d*. Available at: https://free3d.com/premium-3d-models/767 (Accessed: 23 April 2026).

[3] Www.ge.com*/aviation CF6-80C2 high-bypass turbofan engines*. Available at: https://www.geaerospace.com/sites/default/files/datasheet-CF6-80C2.pdf (Accessed: 23 April 2026).

APPENDIX

Table 4: Engine specifications

Parameter	Values
Weight (kg)	*4144*
Static sea level thrust (N)	*233531*
Turbine inlet temperature	*1608 K*
Overall pressure ratio	*30.4*
Bypass ratio	*5.2*
Cruise thrust (N)	*50398*
Air flowrate at sea level (kg/s)	*802.4*